

Functional Annotation Report: PP_2800 (UniProt Q88J49)

Organism: *Pseudomonas putida* KT2440 (strain ATCC 47054 / DSM 6125 / NCIMB 11950) **Locus:** PP_2800 · **Gene product:** Diaminobutyrate–2-oxoglutarate transaminase (putative) **Protein:** 417 aa, ~45.25 kDa · **UniProt:** Q88J49 (unreviewed, TrEMBL) · **RefSeq:** WP_010953709.1 EC / **KO:** EC 2.6.1.76 / KEGG K00836

1. Summary (answer to the research question)

PP_2800 encodes a **soluble, cytoplasmic, pyridoxal-5'-phosphate (PLP)-dependent class-III (ω)-aminotransferase of the DAT/EctB family** (diaminobutyrate–2-oxoglutarate transaminase, EC 2.6.1.76). Enzymes of this family catalyze the **reversible transfer of an amino group between the distal (ω) position of a C4 amino-acid semialdehyde and 2-oxoglutarate/glutamate**, interconverting **L-aspartate- β -semialdehyde + L-glutamate $\rightleftharpoons$ L-2,4-diaminobutyrate (DABA) + 2-oxoglutarate**. This assignment is based on **sequence homology and conserved-domain signatures only** — there is no direct experimental characterization of the PP_2800 protein. Importantly, although the DAT/EctB reaction is the canonical first, rate-limiting step of **ectoine** biosynthesis, *P. putida* KT2440 **lacks the downstream ectoine genes (ectA, ectC, ectD)** and therefore cannot make ectoine. The genome encodes **two K00836 paralogs (PP_2800 and PP_4223, ~51 % identical) that are functionally partitioned**: PP_4223 lies inside the pyoverdine (ferribactin-synthase) NRPS cluster and is the parsimonious supplier of L-2,4-diaminobutyrate (L-Dab) for the pyoverdine siderophore, whereas **PP_2800** sits in a separate GABA/diamine metabolism cluster. Thus the physiological role of PP_2800 is best interpreted as **cytoplasmic metabolism/interconversion of C4 amino compounds (aspartate- β -semialdehyde/DABA-, and possibly GABA-type ω -amino substrates)**, rather than siderophore or compatible-solute synthesis.

2. Molecular identity and evidence

Domain/family evidence (bioinformatic). UniProt and InterPro assign PP_2800 to: - Pfam **PF00202 Aminotran_3** (class-III aminotransferase); PROSITE **PS00600** (class-III PLP AA-transferase active-site motif). - TIGRFAM **TIGR00709 "dat"** and NCBIfam **NF006733 / PRK09264**, both specific to 2,4-diaminobutyrate:2-oxoglutarate 4-aminotransferases. - PANTHER **PTHR43552:SF2 "diaminobutyrate-2-oxoglutarate transaminase."** - InterPro: IPR004637 (Dat), IPR005814/IPR049704 (Aminotrans_3), IPR015421/15424 (PLP-dependent transferase, major/fold). - eggNOG **COG0160** (4-aminobutyrate aminotransferase and related aminotransferases). - Cofactor: **pyridoxal-5'-phosphate** (GO:0030170, PLP binding).

Confidence level in UniProt is **PE 3 "Inferred from homology"** — i.e., predicted, not experimentally verified for this protein.

The catalyzed reaction (from characterized family members). The founding biochemical study of the *Acinetobacter baumannii* Dat enzyme purified the protein and demonstrated the reversible reaction "*L*-2,4-diaminobutyric acid + 2-ketoglutaric acid $\rightleftharpoons$ *L*-glutamic acid + *L*-aspartic beta-semialdehyde" (Ikai & Yamamoto, 1997,).

P 9260954

Recent structural/biochemical work on EctB confirms the physiological direction "*converts aspartate- β -semialdehyde (ASA) to DABA*" and that this is "*the rate-limiting step of ectoine biosynthesis*" (Skogvold et al., 2026,).

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Substrate specificity. DABA aminotransferases are ω -transaminases that **aminate the distal ω -position** of their substrate. Structural work on *Chromohalobacter salexigens* EctB provided "*the first description of two binding pockets in the active site that may account for the catalytic character of DABA aminotransferases*" (Hillier et al., 2020,),

P 32112674

explaining the characteristic **dual-substrate specificity** (one pocket for the amino donor/ glutamate, one for the C4 amino-acceptor). Consistent with broad specificity, the *Marinobacter* EctB showed "*activity was highest using DABA and GABA as substrates*" (

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Such promiscuity means the exact in-vivo substrate of PP_2800 cannot be fixed from homology alone.

3. Cellular localization

PP_2800 is predicted to be a **cytoplasmic (cytosolic) soluble enzyme**. This is inferred from: (i) it is a PLP-dependent central-metabolism aminotransferase, a class that is universally cytosolic; (ii) the sequence begins **MNKIETFERL...** with **no Sec/Tat signal peptide and no transmembrane segment**; and (iii) characterized homologs (EctB, Dat) are soluble oligomeric proteins purified from the cytosol. Family members assemble as **homotetramers built from two functional dimers**, with a "gating loop" from the neighboring monomer helping to shape each active site (P 32112674).

4. Pathway context and the KT2440-specific interpretation

Canonical pathway (in ectoine producers). DABA aminotransferase (EctB) performs the first committed step of ectoine biosynthesis: ASA → DABA; DABA is then N-acetylated by **EctA** and cyclized by **EctC (ectoine synthase)** to ectoine, optionally hydroxylated by **EctD** to 5-hydroxyectoine

(P 32112674);
(P 38960226).

The same Dat reaction also feeds, together with a DABA **decarboxylase (Ddc)**, the **1,3-diaminopropane** pathway in *Acinetobacter* and *Haemophilus*

(P 9260954);
(P 9514614),

and can participate in promiscuous routes such as a "*β*-alanine biosynthetic pathway consisting of promiscuous metabolic enzymes ... including 2,4-diaminobutyrate aminotransferase (Dat) and 2,4-diaminobutyrate decarboxylase (Ddc)" (Perchat et al., 2022,

(P 35623386).

Why the canonical roles do not apply directly to KT2440. A KEGG genome-wide orthology check shows *P. putida* KT2440 encodes the DABA-aminotransferase ortholog K00836 as **two paralogs (PP_2800 and PP_4223)** but has **no ectA (K06718), no ectC (K06720), and no ectD**

(K10674). The ectoine pathway is therefore **genetically incomplete**, and PP_2800 cannot support ectoine synthesis. No DABA decarboxylase (ddc) is present adjacent to PP_2800 either, arguing against a dedicated 1,3-diaminopropane operon at this locus.

Genomic-neighborhood evidence. PP_2800 sits (complement strand, ~3.190–3.192 Mb) within a tight cluster of **C4-amino/ ω -amino-compound metabolism** genes:

Locus	Product	KO / EC
PP_2799	4-aminobutyrate(GABA)–pyruvate transaminase (class III)	K16871 / 2.6.1.96
PP_2800	DABA–2-oxoglutarate transaminase	K00836 / 2.6.1.76
PP_2801	4-aminobutyraldehyde dehydrogenase	K00137 / 1.2.1.19
PP_2802	amino-acid permease	–
PP_2806	TetR-family transcriptional regulator	–

The co-occurrence of a **GABA aminotransferase**, an **aminobutyraldehyde dehydrogenase** (an enzyme of putrescine/polyamine→GABA catabolism), an **amino-acid permease** and a local **TetR regulator** indicates a locus dedicated to **uptake and catabolic interconversion of 4-carbon amino compounds** (GABA/polyamine-derived intermediates). Given the broad ω -transaminase specificity of the family, the most defensible interpretation is that PP_2800 supplies **transamination of aspartate- β -semialdehyde/DABA (and possibly related ω -amino substrates such as GABA)** feeding cytoplasmic amino-acid/polyamine metabolism, rather than a compatible-solute pathway.

4a. The two paralogs are functionally partitioned: PP_4223 (not PP_2800) is the pyoverdine-siderophore enzyme

P. putida KT2440 carries **two K00836 DABA-aminotransferase paralogs, PP_2800 and PP_4223**, which share **50.9 % amino-acid identity** (global alignment, 201/395 aligned positions) and both retain the conserved class-III catalytic-lysine motif (PP_2800 ...ICDEIQAGCGRTGD...TCAK...; PP_4223 ...IVDEIQSGFARTGR...TLSK...). Crucially, their **genomic contexts differ completely**:

- **PP_4223** is embedded **within the pyoverdine (ferribactin) biosynthetic locus** — flanked by PP_4219–PP_4221, the **non-ribosomal peptide synthetase (NRPS) subunits of ferribactin synthase** (ferribactin is the pyoverdine precursor), and PP_4222 (SyrP-like). **L-2,4-diaminobutyrate (L-Dab)** is a documented pyoverdine building block: the

fluorophore is generated "by periplasmic oxidation and cyclization reactions of d-tyrosine and l-diaminobutyric acid" (Ringel et al., 2018,

P 29208656

and the *P. putida* pyoverdine peptide contains an L-Dab residue arising "from the condensation of 2,4-diaminobutyric acid and glutamine" (Gwose & Taraz, 1992,

P 1388514

PP_4223 is therefore the parsimonious source of L-Dab for pyoverdine and the physiological basis of the KEGG "biosynthesis of various siderophores" (ppu00975) mapping of the K00836 orthology.

- **PP_2800**, by contrast, lies **outside any NRPS/siderophore locus**, within the GABA/diamine metabolism cluster described above.

This partition means the siderophore role attributed to the orthology should be assigned primarily to **PP_4223**, whereas **PP_2800 most likely serves general cytoplasmic amino-acid/diamine (C4 ω -amino) metabolism**. The two enzymes may retain overlapping DABA-transaminase activity (potential partial redundancy), but their distinct genomic niches argue for distinct primary roles.

Orthology test (decisive). *P. aeruginosa* PAO1 carries a **single** K00836 gene, **PA2413 = PvdH** ("diaminobutyrate--2-oxoglutarate aminotransferase"), located within its pyoverdine locus and experimentally required to supply L-Dab for pyoverdine. Global pairwise identity shows **PP_4223 is 85.7 % identical to PvdH (382/446)**, i.e. its clear ortholog, while **PP_2800 is only 54.4 % identical to PvdH (211/388)**. This confirms with high confidence that **the pyoverdine DABA-transaminase function in *P. putida* is carried by PP_4223 (the PvdH ortholog), and PP_2800 is a lineage-specific duplication** whose primary role lies outside siderophore biosynthesis, consistent with its GABA/diamine-cluster context.

5. Supported and refuted hypotheses

- **Supported:** PP_2800 is a PLP-dependent class-III DABA-type ω -aminotransferase (EC 2.6.1.76) — strong, converging domain/orthology evidence.
- **Supported:** The protein is cytoplasmic and soluble (no signal peptide/TM; family biochemistry).
- **Refuted (for KT2440):** PP_2800 acts as the EctB step of an ectoine biosynthetic pathway — refuted because ectA/ectC/ectD are absent from the genome.

- **Not supported by local context:** A dedicated 1,3-diaminopropane (dat-ddc) operon at this locus — no adjacent ddc.
- **Reassigned:** The siderophore/pyoverdine role of the K00836 orthology belongs primarily to the paralog **PP_4223** (embedded in the ferribactin-synthase NRPS cluster), not PP_2800.
- **Best current inference:** PP_2800 performs cytoplasmic transamination of C4 amino-semialdehydes (ASA/DABA, and possibly GABA) within an amino-acid/polyamine catabolic/interconversion cluster.

6. Limitations and future directions

- **No direct experimental data exist for PP_2800;** all functional statements are homology/context inferences (UniProt PE3; GO terms IEA only).
 - Class-III ω -transaminases are **catalytically promiscuous**, so the exact preferred substrate and metabolic direction in vivo remain undetermined.
 - The KT2440 genome carries a **paralog, PP_4223 (also K00836, 50.9 % identical)** that is the **PvdH ortholog (85.7 % identical to *P. aeruginosa* PvdH/PA2413)** and supplies pyoverdine L-Dab; whether the two paralogs are functionally redundant under any condition remains to be tested directly.
 - **Recommended experiments:** (i) recombinant expression and steady-state kinetics against ASA, DABA, GABA and putrescine-derived aldehydes; (ii) transcriptional/operon mapping and regulation by the local TetR regulator (PP_2806); (iii) metabolomics of a Δ PP_2800 mutant to detect DABA/GABA/1,3-diaminopropane changes; (iv) an AlphaFold model (available: AlphaFoldDB Q88J49) to confirm the two-pocket active-site architecture and predicted PLP-Lys Schiff-base residue.
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Key references

- Ikai & Yamamoto, 1997, *J Bacteriol* — PMID **9260954** (Dat reaction, EC 2.6.1.76).
- Ikai & Yamamoto, 1998, *J Bacteriol* — PMID **9514614** (dat/ddc, 1,3-diaminopropane).
- Hillier, Altermark & Leiros, 2020, — PMID **32112674** (EctB crystal structure, tetramer, dual active-site pockets).
- Skogvold et al., 2026 — PMID **41652856** (EctB biochemistry; DABA/GABA substrate range).
- Su et al., 2024 — PMID **38960226** (EctB as rate-limiting step; ectABC).

- Perchat et al., 2022 — PMID **35623386** (Dat promiscuity in alternative pathways).
- Ringel, Dräger & Brüser, 2018 — PMID **29208656** (pyoverdine fluorophore formed from D-Tyr + L-Dab; PvdH-type DABA-AT supplies L-Dab).
- Gwose & Taraz, 1992 — PMID **1388514** (*P. putida* pyoverdine structure contains L-Dab).
- Nelson et al., 2002, *Environ Microbiol* — PMID **12534463** (KT2440 genome).

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